CSE237A

Final Project: Submission

One student from each group should submit the following three items to Gradescope on the final demo day.

- Final Project Report
    - Submit a PDF copy of the report named as follows: <userID1>_<userID2>_final_project_report.pdf

- Final Project Slides
    - Submit a PDF copy of the slides. Name it as follows: <userID1>_<userID2>_final_project_slides.pdf
    - There is no limitation on the page length. However, please note that each group will demo the project for a 10-minute time slot. You will present with the slides during the time slot of your demo.

- Final Project Code Upload
    - Upload your code as a zip file with the filename: <userID1>_<userID2>_final_project_code.zip
    - Please put a README file and sufficient comments in your code

Final Project: Final Report

Write a report, single spaced, 11pt Times New Roman font, describing the following:
(Page length: unlimited)

1. Title of your project
2. Team members names, email addresses and student IDs
3. Motivation for your project
4. Related work
    1. Discuss what others have already accomplished in this area - academic papers or commercial products are okay. Include specific citations.
    2. Highlight what is interesting in your particular implementation relative to other previous work
5. List hardware components
    1. Include manufacturer and model of any parts you use
    2. Discuss HW design choices
6. List software components and frameworks
    1. Include release version numbers, etc
    2. Discuss SW design choices
7. Describe how you integrated HW and SW, discuss any issues and challenges you faced and how you solved them
8. Experimental section

    1. Describe your experiment setup and metrics of success (e.g. response delays, target accuracy, energy efficiency)
    2. Present results
    3. Discuss the implications of your results
9. Conclusion
    1. What did you accomplish? How does this demonstrate key concepts in this class or further the development of embedded systems?
    2. Discuss any unfinished parts of the implementations and/or next steps if you were to continue working on the project.

Notes and hints

- The report is a record of your design and experiments, and should include enough detail for someone else to replicate the setup and verify your results. E.g. if you are using the RPi 4 with the custom kernel from this class, you should mention the original kernel version number and customization.
- The report should be a standalone document, i.e. you may reproduce some parts of the motivation/related work sections from your progress report.
- Use pseudocode or flowcharts to describe any algorithms you use, along with descriptive text to explain the overall optimization target. (Do NOT attach C code as an appendix.) Document how you arrived at design parameters like sensor sampling rates, control decision intervals, measurement overheads, etc. **Include units** in any quantitative measurements!
- Negative results or unfinished implementations are acceptable as long as you document the issues (more than just "it didn't work"). Include some speculations for why they happened and give concrete next steps for debugging/solving them if you had more time.